

Disproof of Conjecture 00000000427: Spin-Symmetric Plane Partitions in a Cubic Box at $t = 1$

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Abstract

We disprove Conjecture 00000000427, which asserts that plane partitions in a cubic box that are symmetric under all diagonals and weighted by spin are counted, at $t = 1$, by the closed form $2^{\lfloor n^2/4 \rfloor} \prod_{i=1}^n (2i-1)!!/i!$. At $t = 1$ the spin weight is identically 1, so the claim is purely combinatorial. Complete enumeration of the $n = 1$ and $n = 2$ cases — under two natural readings of “symmetric under all diagonals” — shows the true counts are 2 and 5, whereas the conjectured formula gives 1 and 3. The counterexamples are machine-certified in Lean 4 with zero axioms.

1 Definitions

Definition 1 (Plane partitions in a cubic box). *A plane partition in an $n \times n \times n$ box is an order ideal of the poset $[n] \times [n] \times [n]$. Equivalently, writing the cells of the discrete cube as $\{0, \dots, n-1\}^3$, it is a monotone non-decreasing $\{0, 1\}$ -valued height function*

$$h : \{0, \dots, n-1\}^3 \rightarrow \{0, 1\}, \quad (i \leq i') \wedge (j \leq j') \wedge (k \leq k') \implies h(i, j, k) \leq h(i', j', k').$$

MacMahon’s product formula counts *all* plane partitions in the box:

$$M(n) = \prod_{i,j,k=1}^n \frac{i+j+k-1}{i+j+k-2}, \quad M(1) = 2, \quad M(2) = 20.$$

Definition 2 (Diagonal symmetry). *Let h be a height function on the cube.*

- **Reading A (mirror).** *h is diagonally symmetric if it is invariant under every transposition of coordinates:*

$$h(i, j, k) = h(j, i, k) = h(k, j, i) \quad \text{for all cells } (i, j, k).$$

- **Reading B (cyclic).** *h is diagonally symmetric if it is invariant under the cyclic permutation $(i \rightarrow j \rightarrow k \rightarrow i)$:*

$$h(i, j, k) = h(j, k, i) \quad \text{for all cells } (i, j, k).$$

On the two-dimensional index set $\{0, 1\}^3$ (the $n = 2$ case) both readings generate the same partition of the cube into four orbits

$$\{(0, 0, 0)\}, \quad \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}, \quad \{(1, 1, 0), (1, 0, 1), (0, 1, 1)\}, \quad \{(1, 1, 1)\},$$

so a diagonally symmetric height function is exactly one that is constant on these four orbits, and both readings yield identical counts. We enumerate both anyway, so the disproof below is independent of which reading of “diagonal” is intended.

Definition 3 (Spin weight at $t = 1$). *The conjecture weights symmetric plane partitions by a “spin” parameter t . At $t = 1$ every spin weight equals 1, so spin does not affect any count; the conjecture becomes a purely enumerative statement about Definition 2.*

2 The conjectured formula

Conjecture 00000000427 asserts

$$S(n) = 2^{\lfloor n^2/4 \rfloor} \prod_{i=1}^n \frac{(2i-1)!!}{i!},$$

where $S(n)$ is the number of diagonally symmetric plane partitions in the $n \times n \times n$ box at $t = 1$. Evaluating at small n :

$$F(1) = 2^0 \cdot \frac{1!!}{1!} = 1, \quad F(2) = 2^1 \cdot \frac{1!!}{1!} \cdot \frac{3!!}{2!} = 2 \cdot 1 \cdot \frac{3}{2} = 3.$$

3 Enumeration

We enumerate *all* $\{0, 1\}$ -valued functions on the cube (2^{n^3} of them: 2 for $n = 1$, $2^8 = 256$ for $n = 2$), keep the monotone ones, and among those count the diagonally symmetric ones under each reading of Definition 2. The enumeration is implemented in the accompanying script `reproduce.py` and independently re-verified in Lean 4 (kernel `decide`, zero axioms; see the `lean4/` directory).

n	all functions	monotone	mono. + mirror (A)	mono. + cyclic (B)	conjectured $F(n)$
1	2	2	2	2	1
2	256	20	5	5	3

The monotone counts reproduce MacMahon’s totals $M(1) = 2$ and $M(2) = 20$ exactly, confirming the encoding. For $n = 2$, a diagonally symmetric monotone height function is determined by its four orbit values $(a, b, c, d) = (h(0, 0, 0), h(1, 0, 0), h(1, 1, 0), h(1, 1, 1))$, and monotonicity forces $a \leq b \leq c \leq d$; the five possibilities are

$$(0, 0, 0, 0), (0, 0, 0, 1), (0, 0, 1, 1), (0, 1, 1, 1), (1, 1, 1, 1),$$

i.e. exactly the non-decreasing 0–1 quadruples. The $n = 1$ cube has a single cell and exactly two plane partitions (empty and full), both trivially symmetric.

4 Verdict

Theorem 1 (Disproof of Conjecture 00000000427). *The conjecture is **FALSE**. Under either reading of “symmetric under all diagonals”:*

$$\begin{aligned} n = 1 : \quad & S(1) = 2 \neq 1 = F(1), \\ n = 2 : \quad & S(2) = 5 \neq 3 = F(2). \end{aligned}$$

Remark 1 (Scope). *MacMahon’s product formula itself is correct and is not contested: our enumeration reproduces $M(1) = 2$, $M(2) = 20$. What is refuted is the claimed closed form for the diagonal-symmetry count at $t = 1$, together with the claimed shape of its ratio to the unsymmetrized MacMahon formula (e.g. at $n = 2$ the true ratio is $5/20 = 1/4$, not $3/20$).*

Remark 2 (Robustness of the encoding). *Under the alternative, non-standard convention in which the height function is 3-valued, $h : \{0,1\}^3 \rightarrow \{0,1,2\}$, the monotone functions number 168 (= plane partitions in a $2 \times 2 \times 2 \times 2$ box by MacMahon’s formula) and the diagonally symmetric ones number 15 under both readings — again different from the conjectured value 3. The disproof is thus independent of that encoding choice as well.*

5 Machine certification

The `lean4/` directory contains a Lean 4 project (toolchain `leanprover/lean4:v4.33.1`, library `Main`) proving:

- `macMahon2`: filtering the $2^8 = 256$ height functions of the $2 \times 2 \times 2$ cube by monotonicity leaves exactly 20;
- `symCount2`: monotone + mirror-symmetric count = 5;
- `cycSymCount2`: monotone + cyclic-symmetric count = 5;
- `formula2`: the conjectured value $2^1 \cdot (1!/1!) \cdot (3!/2!) = 3$ (stated as the exact cross-multiplied computation $2^{\lfloor 4/4 \rfloor} \cdot (1 \cdot 3)/(1 \cdot 2) = 3$ over $\mathbb{N}$, since core Lean’s $\mathbb{Q}$ operations do not kernel-reduce);
- `refute2`: the conjunction of the above with `symCount2` $\neq 3$ and `cycSymCount2` $\neq 3$;
- `macMahon1`, `symCount1`, `formula1`, `refute1`: the $n = 1$ counterexample $2 \neq 1$.

All proofs are by kernel `decide/rfl`; `Check.lean` prints `#print axioms` for every theorem, each of which reports that it depends on *no axioms*. There is no `sorry` anywhere.